Mid

Linear Perceptron:

Decision unit to decide whether I will attend a concert or not

$$X_1 = Weather\ (1 = Good\ ,\ 0 = Bad)$$

$$X_2 = Friends\ (\ 1 = With,\ 0 = Without)$$

$$X_3 = Distance\ (1 = Near,\ 0 = Far)$$

$$W_1 = 0.7$$

$$W_2 = 0.3$$

$$W_3 = 0.2$$

$$W_0 + W_1X_1 + W_2X_2 \gtrless 0$$

$$To\ get\ into\ a\ vector\ form:$$

$$\underline{W} = \begin{bmatrix} W_1 \\ W_2 \\ W_3 \\ \dots \\ W_N \end{bmatrix}_{N\times 1},\ \underline{W}^T = \begin{bmatrix} W_1 & W_2 & W_3 & \dots & W_N \end{bmatrix}_{1\times N},\ \underline{X} = \begin{bmatrix} X_1 \\ X_2 \\ X_3 \\ \dots \\ X_N \end{bmatrix}_{N\times 1}$$

$$y = f\left(\underline{W}^T\underline{X} + w_0\right)$$

$$y = sigmoid\left(\underline{W}^T\underline{X} + w_0\right)$$
$$y = tanh\left(\underline{W}^T\underline{X} + w_0\right)$$

$$f\ is\ activation\ function = step\ function$$

Example: Design a decision node (linear perceptron) unit that acts as an AND gate.

X1	X2	Y
0	0	0
0	1	0
1	0	0
1	1	1

Two inputs: (X1 and X2)

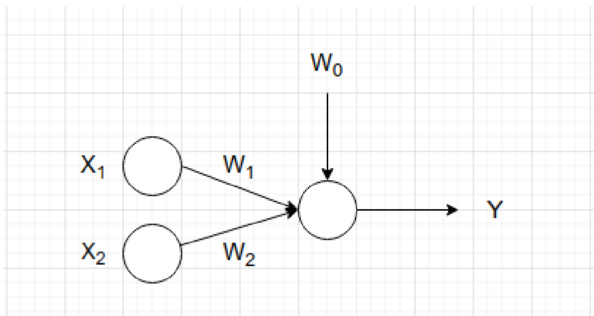


Figure 1: Linear Perceptron for AND

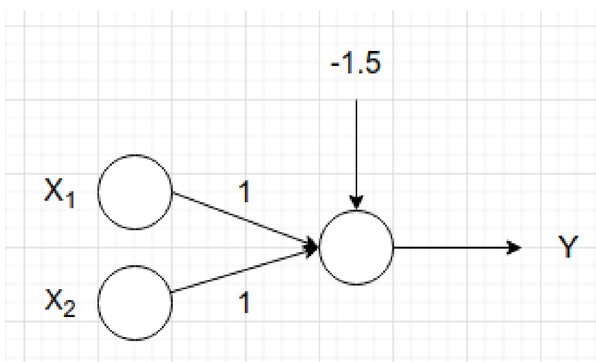
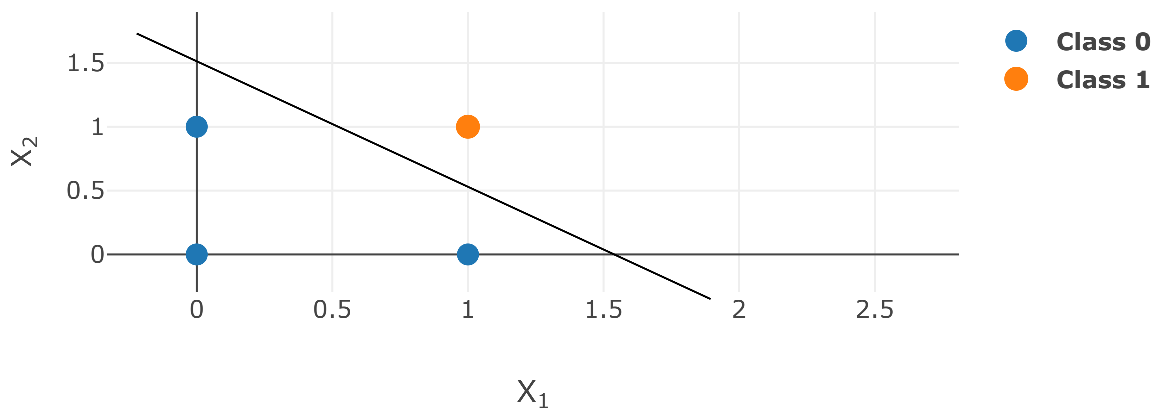


Figure 2:

$$X_1 + X_2 - 1.4 = 0$$



$$slope = \frac{1.5}{-1.5} = -1$$

$$X_2 = -X_1 + 1.5$$

$$X_1 + X_2 - 1.5 = 0$$

$$W_1X_1 + W_2X_2 + W_0 = 0$$

$$W_1 = 1$$
$$W_2 = 1$$
$$W_0 = -1.5$$

Example: Design a decision node (linear perceptron) unit that acts as an OR gate.

X1	X2	Y
0	0	0
0	1	1
1	0	1
1	1	1

Two inputs: (X1 and X2)

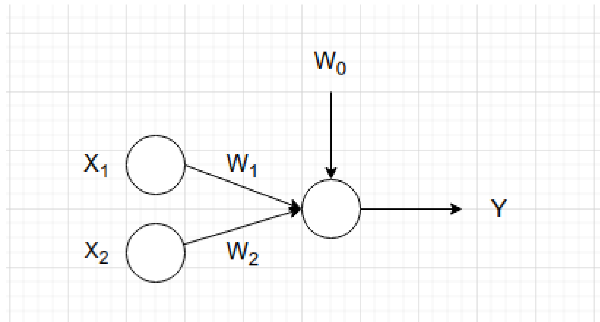


Figure 3: Linear Perceptron for OR

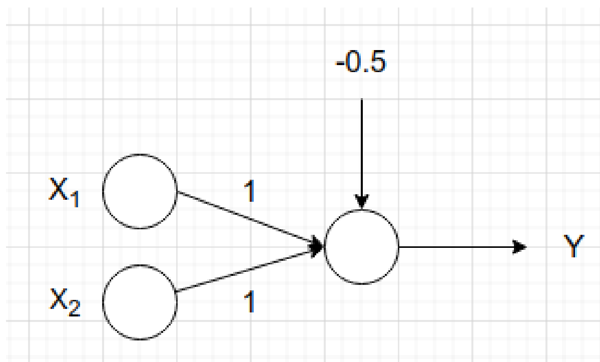
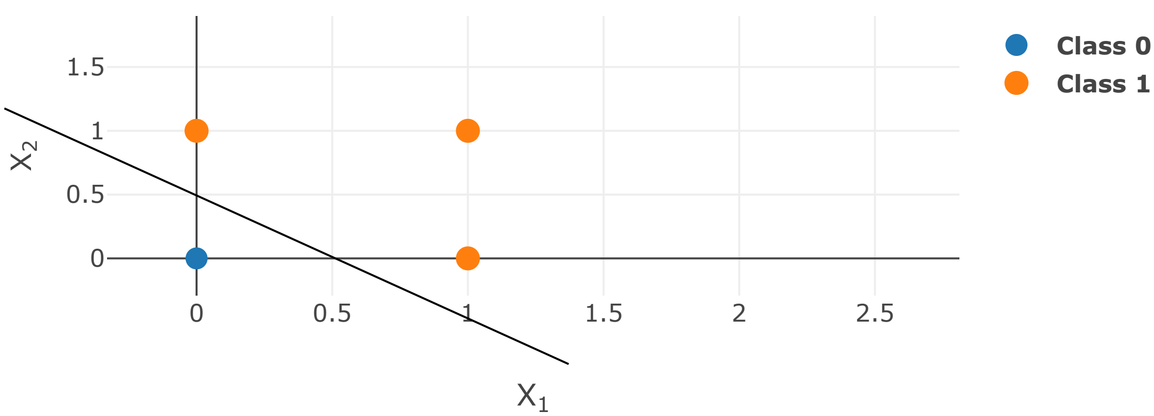


Figure 4:

$$X_1 + X_2 - 0.5 = 0$$



$$slope = \frac{0.5}{-0.5} = -1$$

$$X_2 = -X_1 + c$$

$$X_1 + X_2 - 0.5 = 0$$

$$W_1X_1 + W_2X_2 + W_0 = 0$$

$$W_1 = 1$$
$$W_2 = 1$$
$$W_0 = -0.5$$

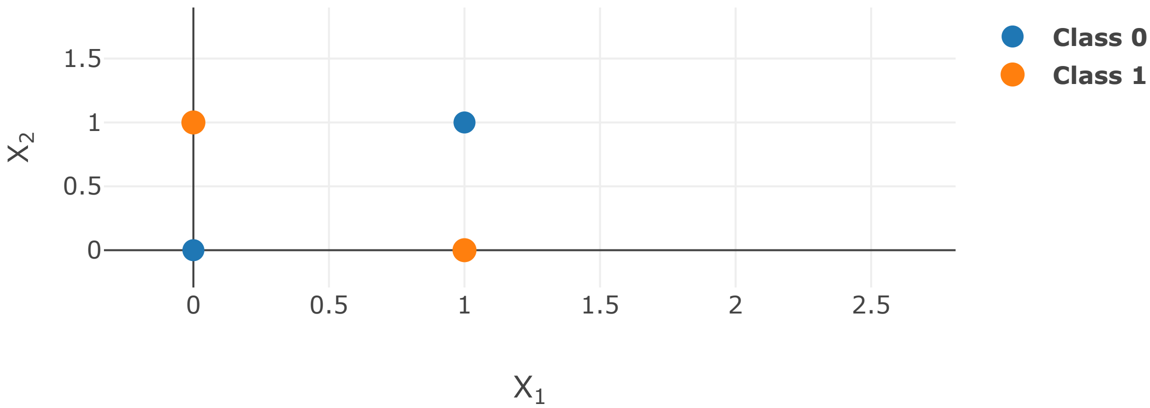
<u>To be solved:</u>
Example: Design a decision node (linear perceptron) unit that acts as an XOR gate.

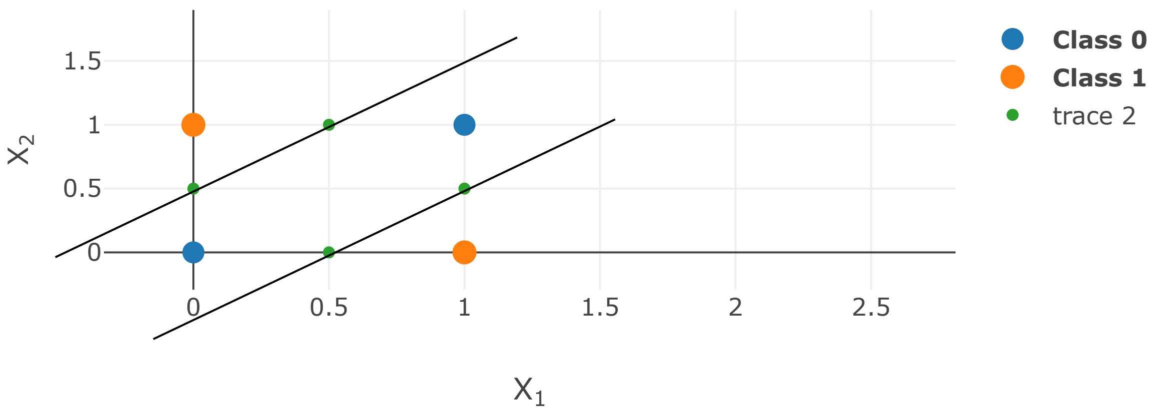
X1	X2	Z1	Z2	Y
0	0	0	1	0
0	1	0	0	1
1	0	1	1	1
1	1	0	1	0

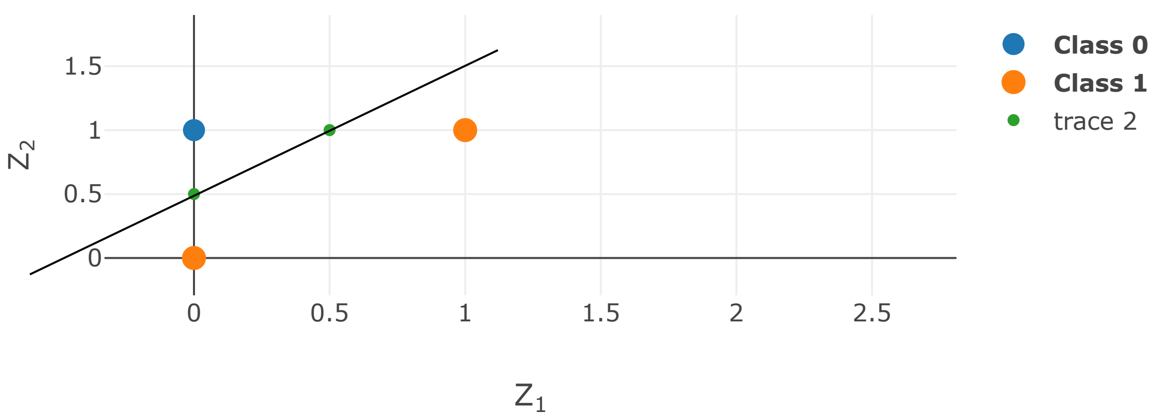
$$Z_1 = f(2X_1 - 2X_2 - 1) \text{ where } f \text{ is a unit step function}$$
$$Z_2 = f(2X_1 - 2X_2 + 1)$$

$$Y = f(Z_1 - Z_2 + 0.5)$$

Two inputs: (X1 and X2)



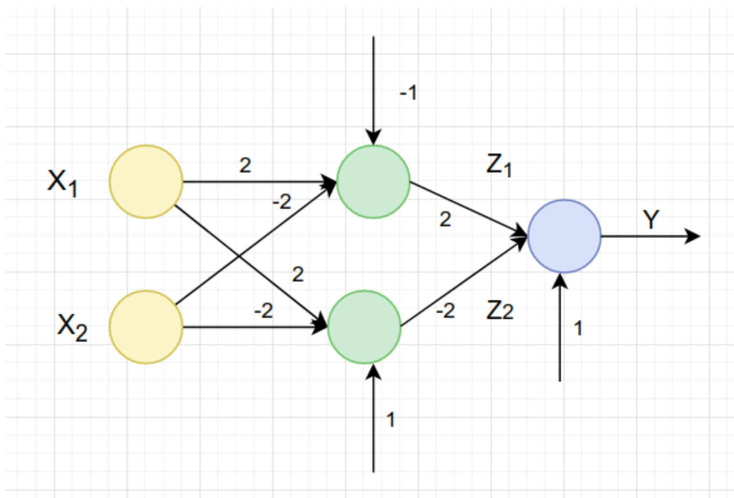




$$\begin{gathered} Equation\ of\ first\ line: \\ X_1 - X_2 - 0.5 = 0 \\ 2X_1 - 2X_2 - 1 = 0 \end{gathered} \tag{1}$$

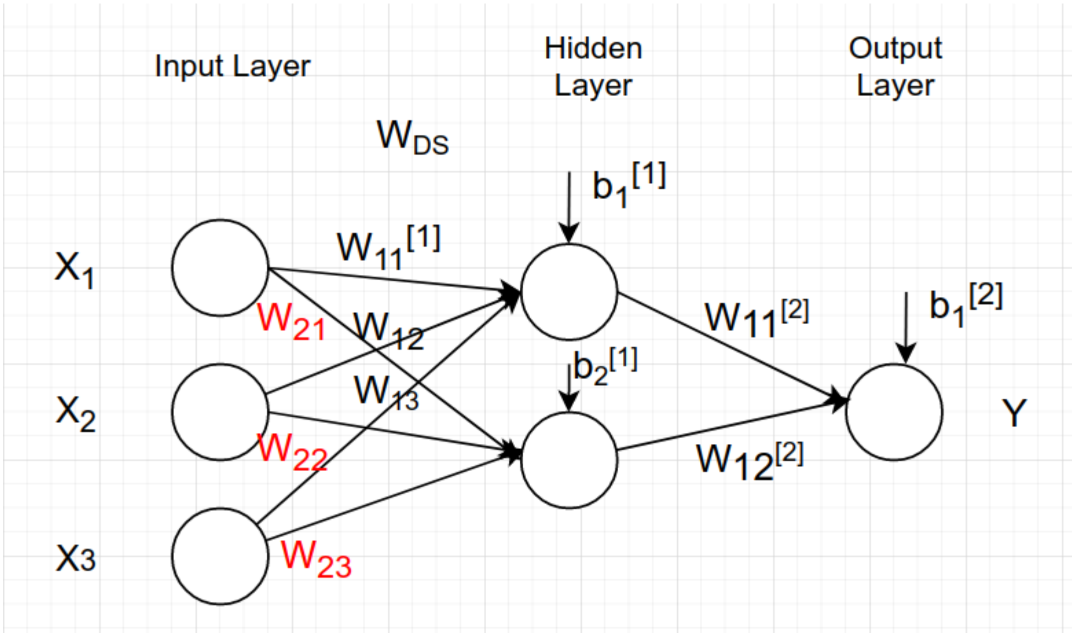
$$\textit{Equation of second line:}$$
$$X_1 - X_2 + 0.5 = 0$$
$$2X_1 - 2X_2 + 1 = 0 \tag{2}$$

$$\textit{Equation of third line:}$$
$$Z_1 - Z_2 + 0.5 = 0$$
$$2Z_1 - 2Z_2 + 1 = 0 \tag{3}$$



Multi-layer Perceptron (MLP)

NEURAL NETWORK



Forward Propagation:

$$z_1^{[1]} = w_{11}^{[1]}\,x_1 + w_{12}^{[1]}\,x_2 + w_{13}^{[1]}\,x_3 + b_1^{[1]}$$
$$a_1^{[1]} = f\left(z_1^{[1]}\right)$$

$$z_2^{[1]} = w_{21}^{[1]}\,x_1 + w_{22}^{[1]}\,x_2 + w_{23}^{[1]}\,x_3 + b_2^{[1]}$$
$$a_2^{[1]} = f\left(z_2^{[1]}\right)$$

$$z_1^{[2]} = w_{11}^{[2]}\,a_1^{[1]} + w_{12}^{[2]}\,a_2^{[1]} + b_1^{[2]}$$
$$y = a_1^{[2]} = f\left(z_1^{[2]}\right)$$

To transform the previous equation into matrix form:

$$\begin{bmatrix} z_1^{[1]} \\ z_2^{[1]} \end{bmatrix} = \begin{bmatrix} w_{11}^{[1]} & w_{12}^{[1]} & w_{13}^{[1]} \\ w_{21}^{[1]} & w_{22}^{[1]} & w_{23}^{[1]} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} + \begin{bmatrix} b_1^{[1]} \\ b_2^{[1]} \end{bmatrix}$$

$$\underline{a}^{[1]} = \begin{bmatrix} f\left(z_1^{[1]}\right) \\ f\left(z_2^{[1]}\right) \end{bmatrix}$$

$$\left[z_1^{[2]} \right] = \left[w_{11}^{[2]} \;\; w_{12}^{[2]} \right] \begin{bmatrix} a_1^{[1]} \\ a_1^{[1]} \end{bmatrix} + \left[b_1^{[2]} \right]$$

$$\underline{z}^{[1]} = \mathbf{W}^{[1]} \underline{a}^{[0]} + \underline{b}^{[1]} \quad // \; matrix\ multiplication$$
$$\underline{a}^{[1]} = f\left(\underline{z}^{[1]}\right) \quad // \, element - wise$$

$$\underline{z}^{[2]} = \mathbf{W}^{[2]} \underline{a}^{[1]} + \underline{b}^{[2]} \quad // \; matrix\ multiplication$$
$$y = \underline{a}^{[2]} = f\left(\underline{z}^{[2]}\right) \quad // \, element - wise$$

In general for any layer l

$$\underline{z}^{[l]} = \mathbf{W}^{[l]} \underline{a}^{[l-1]} + \underline{b}^{[l]} \quad // \; matrix\ multiplication$$
$$\underline{a}^{[l]} = f\left(\underline{z}^{[l]}\right) \quad // \, element - wise$$

$$Dimensions\ of\ \underline{z}^{[l]} = N^{[l]} \times 1$$

$$Dimensions\ of\ \mathbf{W}^{[l]} = N^{[l]} \times N^{[l-1]}$$

$$Dimensions\ of\ \underline{a}^{[l-1]} = N^{[l-1]} \times 1$$

$$Dimensions\ of\ \underline{b}^{[1]} = N^{[l]} \times 1$$

(1) Forward Propagation Equations:

$$\underline{z}^{[1]} = \mathbf{W}^{[1]} \underline{a}^{[0]} + \underline{b}^{[1]} \quad // \; matrix\ multiplication \tag{4}$$
$$\underline{a}^{[1]} = sigmoid\left(\underline{z}^{[1]}\right) \quad // \, element - wise \tag{5}$$

$$\underline{z}^{[2]} = \mathbf{W}^{[2]} \underline{a}^{[1]} + \underline{b}^{[2]} \quad // \; matrix\ multiplication \tag{6}$$
$$\widehat{y} = \underline{a}^{[2]} = sigmoid\left(\underline{z}^{[2]}\right) \quad // \, element - wise \tag{7}$$

(2) Compute cost function: (Cross Entropy Loss Function)

$$\mathcal{L} = -\left[y \log a^{[L\}} + (1-y) \log \left(1 - a^{[L]}\right) \right] \rightarrow L\ stands\ for\ final\ layer \tag{8}$$

$$\mathcal{L} = -[\, y \log \hat{y} + (1-y) \log (1-\hat{y})] \tag{9}$$

$$\mathcal{L} = - \begin{cases} \log \hat{y} & , \quad y = 1 \\ \log (1-\hat{y}) & , \quad y = 0 \end{cases} \tag{10}$$

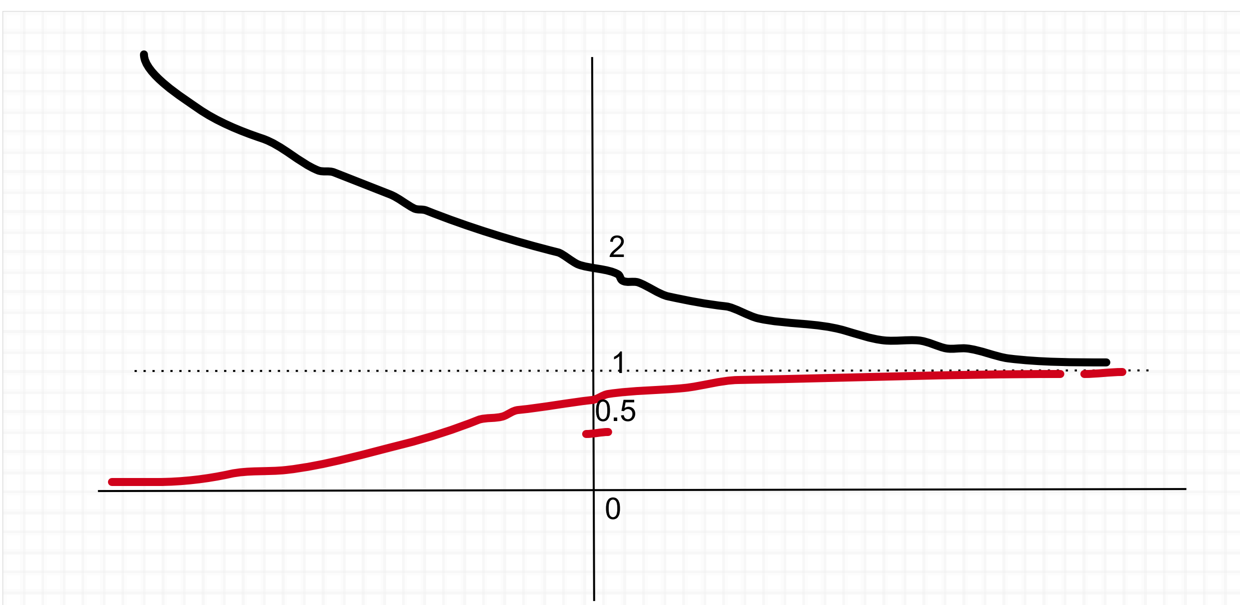
$$y = sigmoid\,(x) = \sigma(x) = \frac{1}{1+e^{-x}} \tag{11}$$

$$\frac{dy}{dx} = \frac{1+e^{-x} - 1}{\left(1+e^{-x}\right)^2} = \frac{1+e^{-x}}{\left(1+e^{-x}\right)^2} - \frac{1}{\left(1+e^{-x}\right)^2} = \frac{1}{1+e^{-x}} - \frac{1}{\left(1+e^{-x}\right)^2}$$

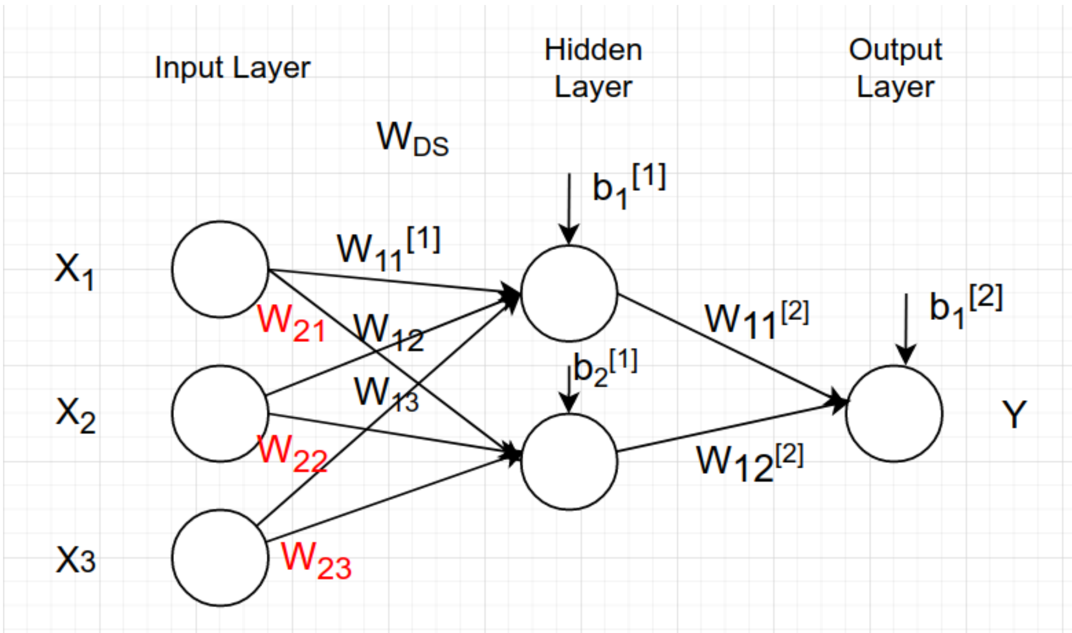
$$= y - y^2 = y(1-y) \tag{12}$$

$$y = tanh(x) = \frac{e^{x} - e^{-x}}{e^{x} + e^{-x}} \tag{13}$$

$$\frac{dy}{dx} = \frac{\left(e^{x}+e^{-x}\right)^2 - \left(e^{x} - e^{-x}\right)^2}{\left(e^{x}+e^{-x}\right)^2} = 1 - y^2 \tag{14}$$



(3) Backpropagation:



- What we have done so far:

$$\underline{z}^{[1]} = \mathbf{W}^{[1]}\underline{a}^{[0]} + \underline{b}^{[1]} \quad // \; matrix\ multiplication \tag{15}$$

$$\underline{a}^{[1]} = sigmoid\left(\underline{z}^{[1]}\right) \quad // \, element - wise \tag{16}$$

$$\underline{z}^{[2]} = \mathbf{W}^{[2]}\underline{a}^{[1]} + \underline{b}^{[2]} \quad // \; matrix\ multiplication \tag{17}$$

$$\hat{y} = \underline{a}^{[2]} = sigmoid\left(\underline{z}^{[2]}\right) \quad // \, element - wise \tag{18}$$

$$\mathfrak{L} = -\left[\; y\ log\ a^{[2\}} + (1-y)\ log\ \left(1 - a^{[2]}\right) \right] \;\rightarrow\; L\ stands\ for\ final\ layer \tag{19}$$

- What we need to do:

$$We\ need\ to\ minimize\ the\ loss\ function\ with\ respect\ to\ W\ and\ b$$
$$= Or\ equivalently\ we\ need\ to\ find\ the\ set\ of\ weights\ and\ biases$$
$$that\ minimizes\ the\ loss\ function\ \mathfrak{L}$$

$$= \; We\ need\ to\ compute\ \frac{\partial L}{\partial \mathbf{W}^{[2]}},\ \frac{\partial L}{\partial \underline{b}^{[2]}},\ \frac{\partial L}{\partial \mathbf{W}^{[1]}},\ \frac{\partial L}{\partial \underline{b}^{[1]}},$$

$$\frac{\partial L}{\partial \mathbf{W}^{[2]}} = \frac{\partial L}{\partial a^{[2]}} * \frac{\partial a^{[2]}}{\partial z^{[2]}} * \frac{\partial z^{[2]}}{\partial \mathbf{W}^{[2]}} \tag{20}$$

$$\frac{\partial L}{\partial b^{[2]}} = \frac{\partial L}{\partial a^{[2]}} * \frac{\partial a^{[2]}}{\partial z^{[2]}} * \frac{\partial z^{[2]}}{\partial b^{[2]}} \tag{21}$$

$$\frac{\partial L}{\partial \mathbf{W}^{[1]}} = \frac{\partial L}{\partial z^{[2]}} * \frac{\partial z^{[2]}}{\partial a^{[1]}} * \frac{\partial a^{[1]}}{\partial z^{[1]}} * \frac{\partial z^{[1]}}{\partial \mathbf{W}^{[1]}} \tag{22}$$

$$\frac{\partial L}{\partial b^{[1]}} = \frac{\partial L}{\partial z^{[2]}} * \frac{\partial z^{[2]}}{\partial a^{[1]}} * \frac{\partial a^{[1]}}{\partial z^{[1]}} * \frac{\partial z^{[1]}}{\partial b^{[1]}} \tag{23}$$

Forward Propagation:

$$\underline{z}^{[1]} = \mathbf{W}^{[1]}\underline{a}^{[0]} + \underline{b}^{[1]} \quad // \textit{ matrix multiplication}$$

$$\underline{a}^{[1]} = sigmoid\left(\underline{z}^{[1]}\right) \quad // \textit{elementwise}$$

$$\underline{z}^{[2]} = \mathbf{W}^{[2]}\underline{a}^{[1]} + \underline{b}^{[2]} \quad // \textit{ matrix multiplication}$$

$$\hat{y} = \underline{a}^{[2]} = sigmoid\left(\underline{z}^{[2]}\right) \quad // \textit{elementwise}$$

$$\mathfrak{L} = -\left[y \log a^{[2\}} + (1-y) \log\left(1 - a^{[2]}\right)\right]$$

Backward Propagation:

$$\frac{\partial L}{\partial \mathbf{W}^{[2]}} = \frac{\partial L}{\partial a^{[2]}} * \frac{\partial a^{[2]}}{\partial z^{[2]}} * \frac{\partial z^{[2]}}{\partial \mathbf{W}^{[2]}}$$

$$\frac{\partial L}{\partial b^{[2]}} = \frac{\partial L}{\partial a^{[2]}} * \frac{\partial a^{[2]}}{\partial z^{[2]}} * \frac{\partial z^{[2]}}{\partial b^{[2]}}$$

$$\frac{\partial L}{\partial \mathbf{W}^{[1]}} = \frac{\partial L}{\partial z^{[2]}} * \frac{\partial z^{[2]}}{\partial a^{[1]}} * \frac{\partial a^{[1]}}{\partial z^{[1]}} * \frac{\partial z^{[1]}}{\partial \mathbf{W}^{[1]}}$$

$$\frac{\partial L}{\partial b^{[1]}} = \frac{\partial L}{\partial z^{[2]}} * \frac{\partial z^{[2]}}{\partial a^{[1]}} * \frac{\partial a^{[1]}}{\partial z^{[1]}} * \frac{\partial z^{[1]}}{\partial b^{[1]}}$$

$$da_2 = \frac{\partial L}{\partial a^{[2]}} = -\frac{y}{a^{[2]}} + \frac{1-y}{1-a^{[2]}} = \frac{a^{[2]} - y}{a^{[2]}\left(1 - a^{[2]}\right)} \tag{24}$$

$$dz_2 = \frac{\partial L}{\partial z^{[2]}} = \frac{\partial L}{\partial a^{[2]}} * \frac{\partial a^{[2]}}{\partial z^{[2]}} = \frac{a^{[2]} - y}{a^{[2]}\left(1 - a^{[2]}\right)} * a^{[2]}\left(1 - a^{[2]}\right) = a^{[2]} - y \tag{25}$$

$$(dW_2)_{1\times 2} = \frac{\partial L}{\partial W^{[2]}} = dz_2 * \frac{\partial z^{[2]}}{\partial \mathbf{W}^{[2]}} = \left(a^{[2]} - y\right)_{1\times 1} (\underline{a}^{[1]^T})_{1\times 2} \tag{26}$$

$$(db_2)_{1\times 1} = \frac{\partial L}{\partial b^{[2]}} = dz_2 \frac{\partial z^{[2]}}{\partial b^{[2]}} = \left(a^{[2]} - y\right)_{1\times 1} \tag{27}$$

$$(da_1)_{2\times 1} = \frac{\partial L}{\partial a^{[1]}} = (\, dz_2 \,)_{1\times 1} \frac{\partial z^{[2]}}{\partial a^{[1]}} = \left((\, dz_2 \,)_{1\times 1} \left(\mathbf{W}^{[2]}\right)_{1\times 2}\right)^T \tag{28}$$

$$(dz_1)_{2\times1} = \frac{\partial L}{\partial z^{[1]}} = (da_1)_{2\times1} * \frac{\partial a^{[1]}}{\partial z^{[1]}} = (da_1)_{2\times1} * a^{[1]}\left(1 - a^{[1]}\right) \tag{29}$$

$$(dW_1)_{2\times3} = (dz_1)_{2\times1} \frac{\partial z^{[1]}}{\partial \mathbf{W}^{[1]}} = (dz_1)_{2\times1} \left(a^{[0]^T}\right)_{1\times3} \tag{30}$$

$$(db_1)_{2\times1} = (dz_1)_{2\times1} * \frac{\partial z^{[1]}}{\partial b^{[1]}} = (dz_1)_{2\times1} \tag{31}$$

We are now able to compute each of:

$$\frac{\partial L}{\partial \mathbf{W}^{[2]}}, \frac{\partial L}{\partial b^{[2]}}, \frac{\partial L}{\partial \mathbf{W}^{[1]}}, \frac{\partial L}{\partial b^{[1]}}$$

4. Updating the weights (Gradient Descent)
After each iteration, we should update the weights based on the computed derivatives , and also
based on Widrow-Hoff Update Rule:

α is called learning rate

$$\mathbf{W}^{[2]} := \mathbf{W}^{[2]} - \alpha\frac{\partial L}{\partial \mathbf{W}^{[2]}} \tag{32}$$

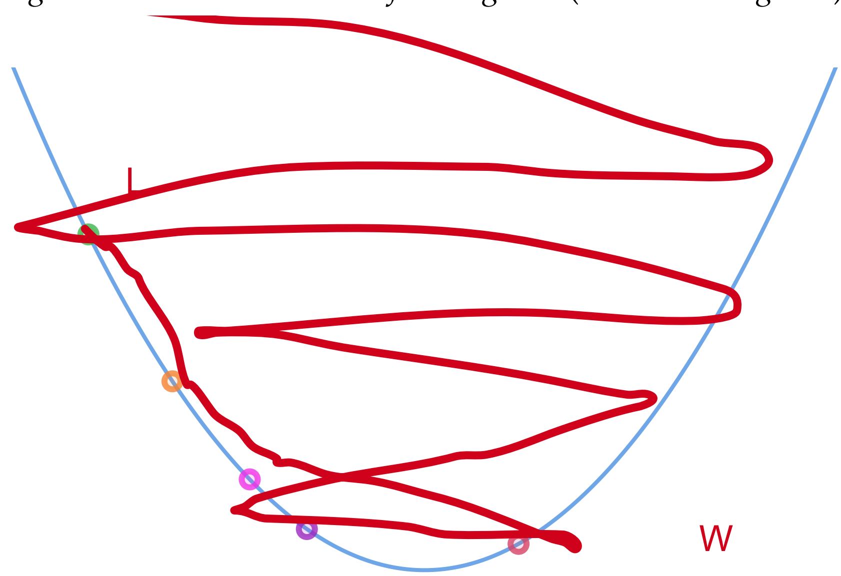
$$b^{[2]} := b^{[2]} - \alpha\frac{\partial L}{\partial b^{[2]}} \tag{33}$$

$$\mathbf{W}^{[1]} := \mathbf{W}^{[1]} - \alpha\frac{\partial L}{\partial \mathbf{W}^{[1]}} \tag{34}$$

$$b^{[1]} := b^{[1]} - \alpha\frac{\partial L}{\partial b^{[1]}} \tag{35}$$

$$\alpha \text{ is too small } \rightarrow \text{ networks takes too long to converge (reach global minimum)}$$

$$\alpha \text{ is too large } \rightarrow \text{ network is at risk of divergence (non} - \text{convergence)}$$



Hyperparameters:

1. Learning Rate
   (a) α = 0.1 , validation accuracy = 0.76
   (b) α = 0.2 , validation accuracy = 0.82
   (c) α = 0.01 , validation accuracy = 0.93
   (d) α = 0.001 , validation accuracy = 0.945
   (e) α = 0.0001 , validation accuracy = 0.932
2. Momentum paramters β_1 *and* β_2

1. Understand the nature of the problem (M = 7)

Training set:

Patient ID	X_1 (Age)	X_2 (Blood sugar level)	X_3 (Body mass index)	Y (Diabetic or not)
1	**35**	**100**	**75**	**0**

2	76	140	78	1
3	42	110	82	1
4	30	98	67	1

Validation Set:

Patient ID	X_1 (Age)	X_2 (Blood sugar level)	X_3 (Body mass index)	Y (Diabetic or not)
5	17	120	90	0.7 (1)
6	56	130	100	0.1 (0)
7	79	134	95	0.3 (0)

Test Set:

Patient ID	X_1 (Age)	X_2 (Blood sugar level)	X_3 (Body mass index)	Y (Diabetic or not)
1	56	119	98	?
2	42	109	78	?
3				
4				
5				
6				
7				

2. Understand the model itself.

* Three layered network. (Input Layer l[0], Hidden layer l[1], Output layer l[2])
* Unknown : $W^{[1]}_{11},\ W^{[1]}_{12},\ \ W^{[1]}_{13},\ W^{[1]}_{21},\ W^{[1]}_{22},\ W^{[1]}_{23},\ b^{[1]}_{1},\ b^{[2]}_{2},\ W^{[2]}_{11},\ W^{[2]}_{12},\ \ b^{[2]}_{1}$

<u>3. How do we compute these weights?</u>

We use our training set to perform the following:

Sequential (Stochastic) Gradient Descent (SGD)

Initialize weights and biases randomly.

for i = 1: N iterations → Training Loop (epochs)

for j = 1: M (M: # examples):

1. **Forward Propagation**
2. **Compute the loss function.**
3. **Backward Propagation.**
4. **Update the weights and biases.**

Batch Gradient Descent (BGD)

Initialize weights and biases randomly.

for i = 1: N iterations → Training Loop (epochs)

for j = 1: M (M: # examples):

1. **Forward Propagation**
2. Accumulate to the loss

3. **Backward Propagation. (with averaging over M)**
4. **Update the weights and biases.**

Mini-Batch Gradient Descent (BGD)

Initialize weights and biases randomly.

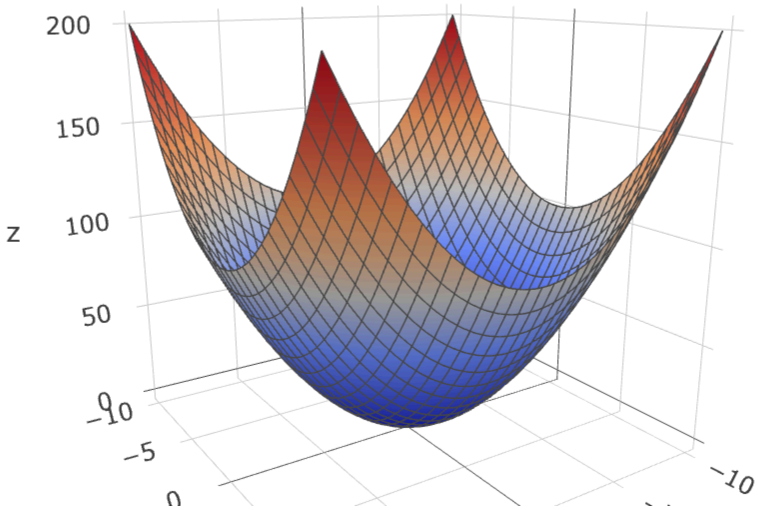
for i = 1: N iterations → Training Loop (epochs)

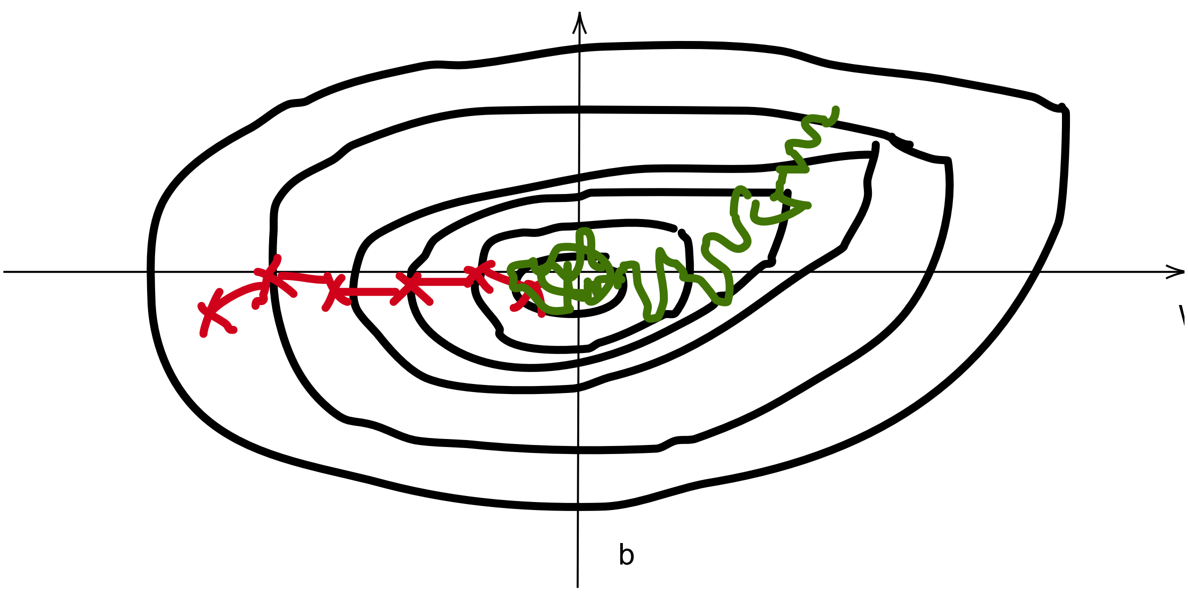
for j=1: minibatches

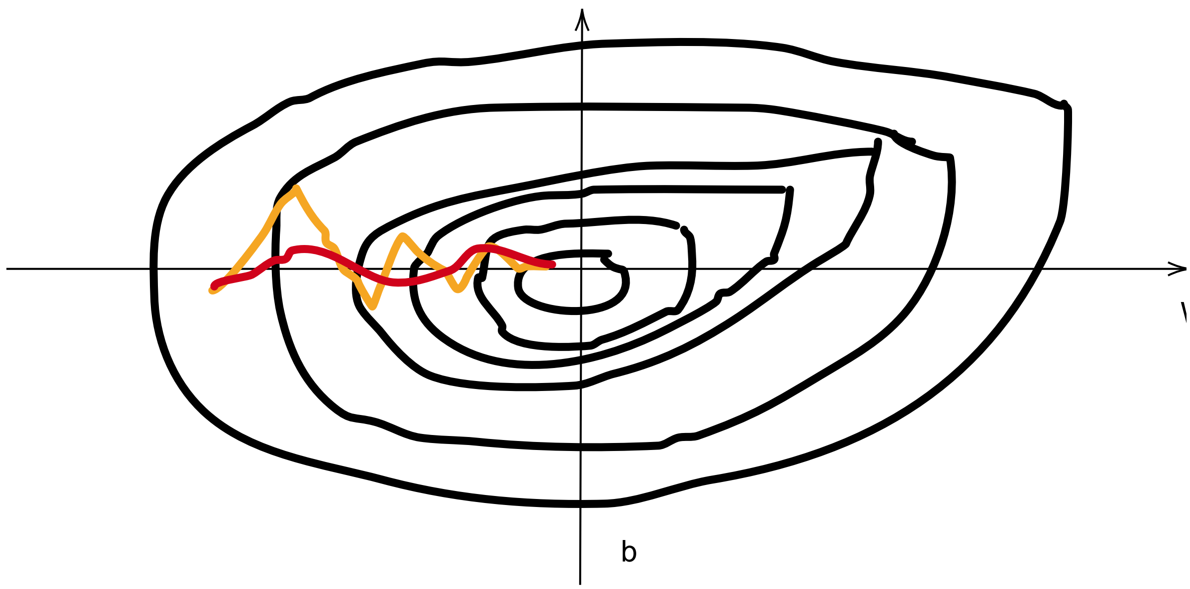
for k = 1: M (M: # examples in the jth minibatch):

1. **Forward Propagation**
2. Accumulate to the loss

3. **Backward Propagation. (with averaging over M)**
4. **Update the weights and biases.**


200
150
z
100
50
0
−10
−5
0
−10


b



SGD with momentum:

$$
\begin{aligned}
dW_0 &= 0 \\
db_0 &= 0
\end{aligned}
$$

$$
dW_t = \frac{\beta dW_{t-1} + (1-\beta)\left(\frac{\partial L}{\partial \mathbf{W}}\right)_t}{1-\beta}
$$

$$
db_t = \frac{\beta\, db_{t-1} + (1-\beta)\left(\frac{\partial L}{\partial b}\right)_t}{1-\beta}
$$

$$
\mathbf{W} := \mathbf{W}^{[2]} - \alpha dW_t \tag{36}
$$

$$
b := b^{[2]} - \alpha db_t \tag{37}
$$

RMSProp:

$$SW_0 = 0$$
$$Sb_0 = 0$$

$$SW_t = \frac{\beta_2 \; SW_{t-1} + (1-\beta_2)\left(\frac{\partial L}{\partial \mathbf{W}}\right)_t^2}{1-\beta_2}$$

$$Sb_t = \frac{\beta \; Sb_{t-1} + (1-\beta_2)\left(\frac{\partial L}{\partial b}\right)_t^2}{1-\beta_2}$$

$$\mathbf{W} := \mathbf{W}^{[2]} - \alpha \frac{\frac{\partial L}{\partial \mathbf{W}}}{\sqrt{S_W + \epsilon}} \tag{38}$$

$$b := b^{[2]} - \alpha \frac{\frac{\partial L}{\partial b}}{\sqrt{S_b + \epsilon}} \tag{39}$$

(40)